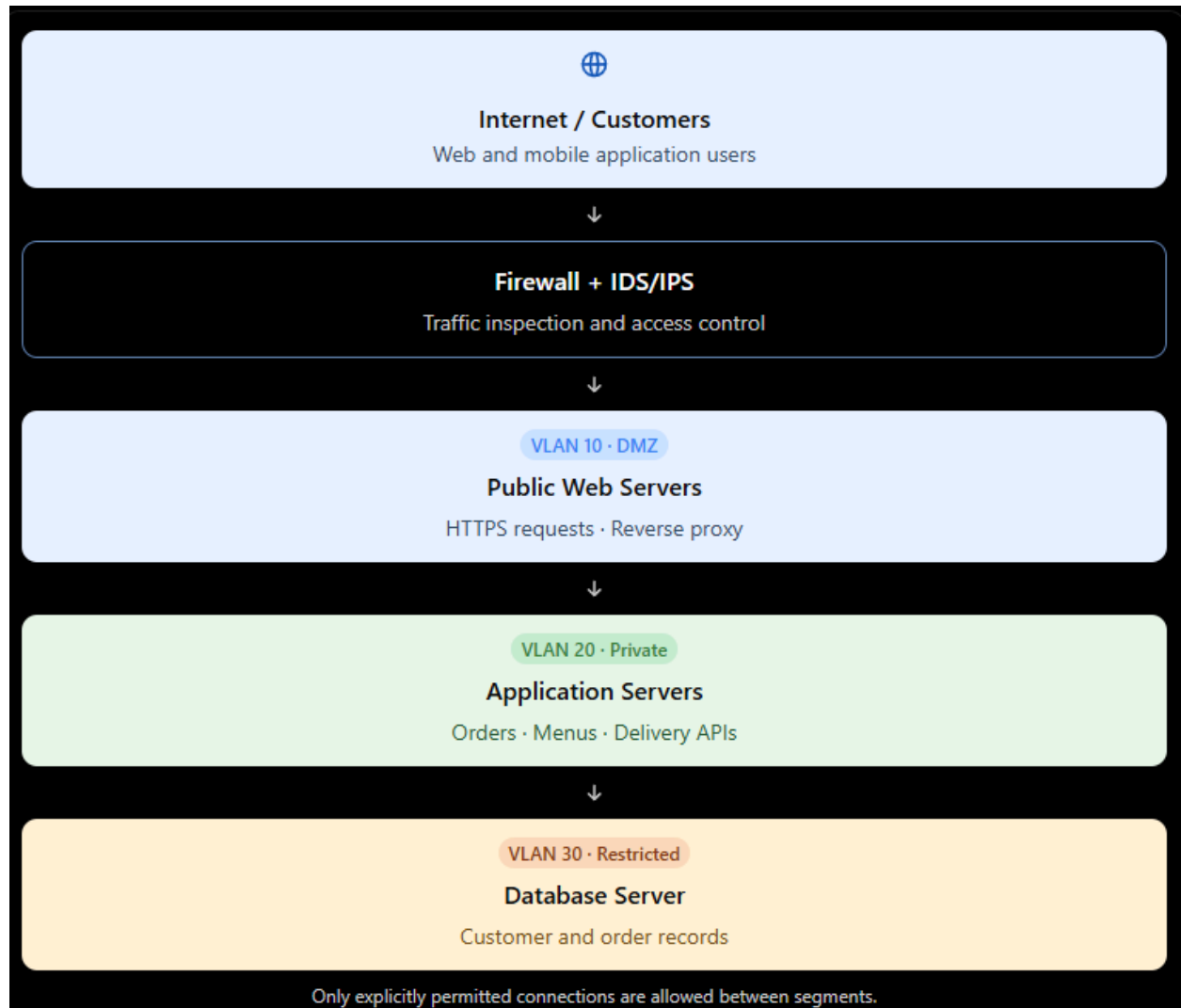


Task 1: Network Diagram for a Food Delivery Application

Objective

To design a segmented network for a food delivery backend with separate public web, application, and database segments.

Network Diagram



Explanation

1. Public web servers receive HTTPS requests from customers.
2. Application servers process orders, user accounts, restaurant information, and delivery requests.

3. The database server stores application data and accepts connections only from authorized application servers.
4. A firewall controls traffic between the segments.
5. An IDS/IPS monitors traffic for suspicious activity.

Task 2: Network Segmentation Plan for an Online Ticket Booking System

Objective

To design a segmented network for an online ticket booking platform using VLANs or subnets.

Proposed Segmentation Table

VLAN / Segment	Devices and services	Access policy
VLAN 10 – Public DMZ	Web servers, reverse proxy	Internet access to HTTPS only
VLAN 20 – Application	Booking APIs, ticket inventory services	Accepts requests from the DMZ
VLAN 30 – Payment	Payment integration services	Restricted application and payment-provider communication
VLAN 40 – Database	Customer, booking and inventory databases	Accessible only by authorized services
VLAN 50 – Management	Admin workstations, monitoring and jump server	Accessible only to authorized administrators

Security Benefits

1. Reduced attack surface: Public-facing systems cannot directly connect to every internal server.
2. Protection of payment services: Payment-related components are isolated from ordinary web services.

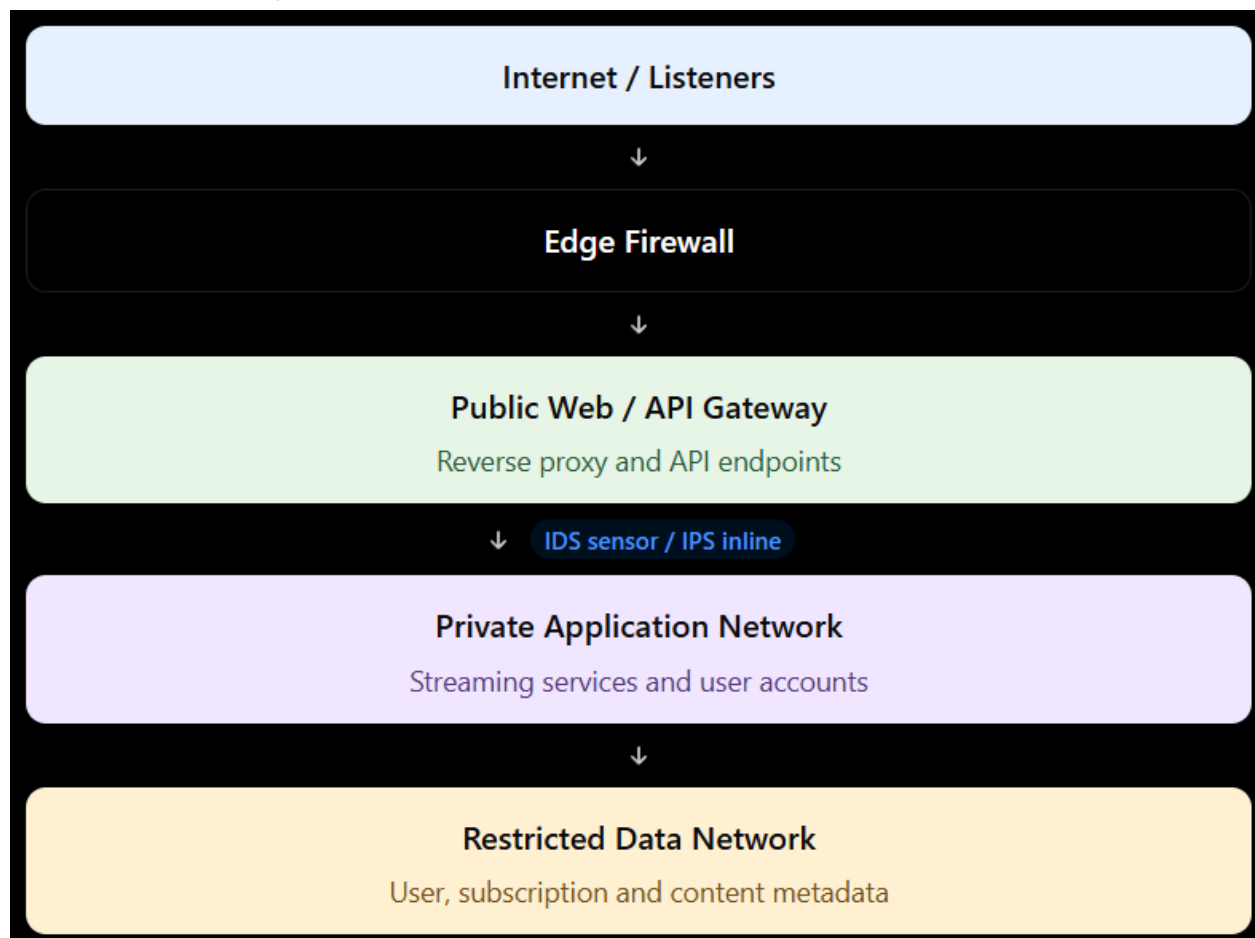
3. Database isolation: Only approved application services can access the database.
4. Better monitoring: Traffic between VLANs can be inspected and logged.
5. Controlled administration: Management access is restricted to authorized administrators.
6. Limiting lateral movement: If one server is compromised, segmentation can prevent an attacker from freely accessing other systems.

Task 3: IDS/IPS Placement for a Music Streaming Service

Objective

To select and place an IDS/IPS in a segmented music streaming network.

Proposed Deployment



Why Deploy Snort Here?

Snort can monitor traffic entering the application network and detect suspicious requests before they reach sensitive backend services. An inline IPS deployment can block matching traffic when configured with appropriate rules and policies.

Attacks to Detect

Attack type	Detection objective
SQL injection	Identify suspicious database-related input
Cross-site scripting (XSS)	Detect common script-injection patterns
Port scanning	Identify reconnaissance activity
Brute-force attempts	Detect repeated authentication requests
Denial-of-service patterns	Identify abnormal connection or request rates
Known exploit attempts	Match traffic against available detection rules

Task 4: Peer Review of a Network Design

Objective

To review a peer's network segmentation and IDS/IPS placement and suggest two improvements.

Sample Peer Design

The peer designed a ticket booking network with public web servers, application servers, and a database server. An IDS was placed at the internet gateway.

Suggested Improvements

Improvement 1: Isolate payment and database services.

Create separate VLANs or subnets for payment processing and database servers. Configure firewall rules so that only approved application services can connect to these systems.

Reason: This limits unauthorized access to payment-related services and sensitive customer information.

Improvement 2: Add internal traffic monitoring.

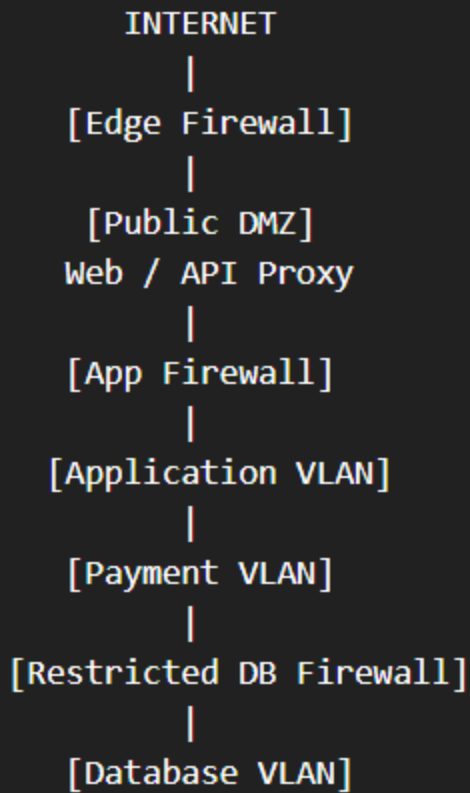
Deploy an additional IDS sensor between the application and database segments. Forward security alerts to a centralized logging system.

Reason: An internet-facing sensor may not see all internal traffic. Monitoring the application-to-database connection provides better visibility into suspicious internal activity.

Task 5: Updated Network Diagram and Review Note

Updated Design

Following the review, the network was updated to include a separate payment VLAN and an IDS sensor between the application and database networks.



IDS Sensor --> Monitors internal traffic

Admin VLAN --> Restricted management access